

1.

a) List four access technologies. Classify each one as home access, enterprise access, or wide-area wireless access.

b) What access network technologies would be most suitable for providing internet access in rural areas?

c) HFC, DSL, and FTTH are all used for residential access. For each of these access technologies, provide a range of transmission rates and comment on whether the transmission rate is shared or dedicated.

a) Four Access Technologies and Their Classifications

1. Digital Subscriber Line (DSL)

Classification: Home Access

- Uses existing telephone lines to provide internet connectivity to residential users
- Connects homes directly to the Internet Service Provider's central office

2. Ethernet

Classification: Enterprise Access

- High-speed wired connection commonly used in office buildings and corporate networks
- Provides reliable, high-bandwidth connectivity for business environments

3. 4G/5G Cellular Networks

Classification: Wide-area Wireless Access

- Provides internet connectivity over large geographical areas using cellular towers
- Enables mobile internet access across cities, regions, and countries

4. Cable Internet (HFC - Hybrid Fiber Coaxial)

Classification: Home Access

- Uses cable TV infrastructure to deliver internet services to residential customers
- Leverages existing coaxial cable networks in neighborhoods

b) Suitable Access Network Technologies for Rural Areas

In rural areas, it is difficult and costly to lay cables over long distances. So, wireless and simple technologies are more suitable. The best options include:

1. **Wireless Technologies (like 4G LTE, 5G, or Fixed Wireless)** – These use mobile towers to provide internet without needing underground cables. They are fast and easy to set up, making them ideal for remote areas.
2. **Satellite Internet** – This uses a satellite dish to receive internet from satellites in the sky. It works well in very remote villages where there are no mobile towers or cables. Though it's slower and more expensive, it gives coverage everywhere.
3. **DSL (Digital Subscriber Line)** – If telephone lines are already available in the area, DSL can provide basic internet. It is slower than other modern methods but still better than no connection.
4. **Community Wi-Fi** – This is a local wireless network shared by many people in a small area. A central internet connection is distributed using Wi-Fi signals. It is low-cost and good for small rural communities.

c) HFC, DSL, and FTTH: Speed and Sharing

Meanings:

- **HFC (Hybrid Fiber-Coaxial)**: This uses both fiber optic and coaxial TV cables to give internet to homes.
- **DSL (Digital Subscriber Line)**: This gives internet through regular telephone lines.
- **FTTH (Fiber To The Home)**: This uses pure fiber optic cable directly to your house for very fast internet.

These are technologies used at home to get internet:

Technology	Speed Range	Shared or Dedicated?
HFC	• Downstream: Up to 1.2 Gbps. • Upstream: 30–100 Mbps	Shared – Speed may slow if many users nearby use it together.
DSL	- o Downstream: 24–52 Mbps. - o Upstream: 3.5–16 Mbps	Dedicated – Each house has its own line, but speed depends on distance.
FTTH	100 Mbps – 1 Gbps or more	Dedicated – Fiber line directly to your home gives high and stable speed.



- **Shared** = Many people use the same network, so speed may drop.
 - **Dedicated** = Only you use it, so speed is more stable.
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2. Suppose users share a 2 Mbps link. Also suppose each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20 percent of the time. (See the discussion of statistical multiplexing in Section 1.3.)

- a) When circuit switching is used, how many users can be supported?
- b) For the remainder of this problem, suppose packet switching is used. Why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time? Why will there be a queuing delay if three users transmit at the same time?
- c) Find the probability that a given user is transmitting.
- d) Suppose now there are three users. Find the probability that at any given time, all three users are transmitting simultaneously. Find the fraction of time during which the queue grows.

Given:

- Link capacity = **2 Mbps**
- Each user transmits at **1 Mbps** when active
- Each user is **active (transmitting)** only **20% of the time** = 0.2
- Remaining **80% of the time** = inactive
- This is an example of **statistical multiplexing** (from Section 1.3)

(a) When circuit switching is used, how many users can be supported?

Circuit switching is primarily used in situations requiring a dedicated, continuous communication path between two endpoints, particularly for voice calls and some data transmissions.

In **circuit switching**, resources are reserved in advance, and each user gets a **fixed bandwidth**.

- Each user requires **1 Mbps** when active.
- Total link capacity = **2 Mbps**.

Number of supported users:

$$\text{Max users} = \left\lfloor \frac{\text{Link capacity}}{\text{User rate}} \right\rfloor = \left\lfloor \frac{2 \text{ Mbps}}{1 \text{ Mbps}} \right\rfloor = 2 \text{ users}$$

(b) In packet switching:

- Why **no queuing delay** if **2 or fewer users** transmit at the same time?
- Why **queuing delay** if **3 users** transmit at the same time?

Answer (b):

- If **2 users** transmit at the same time:
 - They use **$2 \times 1 \text{ Mbps} = 2 \text{ Mbps}$**
 - This matches the link's total capacity
 - So, all data can be sent immediately, and there is **no waiting**, i.e., **no queuing delay**
- If **3 users** transmit at the same time:

- They try to use **$3 \times 1 \text{ Mbps} = 3 \text{ Mbps}$**
- But the link can only handle 2 Mbps
- So, **1 Mbps worth of data has to wait**, causing a **queuing delay**

Summary:

No delay if the total sending rate $\leq 2 \text{ Mbps}$.

Delay happens when the sending rate $> 2 \text{ Mbps}$.

(c) What is the probability that a given user is transmitting?

The problem says each user transmits **20% of the time**, so:

Answer (c): Probability = **0.2** (or **20%**)

(d) Suppose there are 3 users.

We want:

- The **probability that all 3 users** are transmitting at the same time.
- The **fraction of time** during which the **queue grows**.

Each user transmits independently with probability = 0.2

To find the probability all 3 transmit at the same time:

$$P(\text{All 3 transmit}) = 0.2 \times 0.2 \times 0.2 = 0.008$$

So, **0.8%** of the time, all 3 users transmit at the same time.

Fraction of time the queue grows:

The queue grows when the arrival rate exceeds the service rate, which happens only when all 3 users transmit simultaneously (as we determined in part b).

Therefore, the fraction of time during which the queue grows = $P(\text{all 3 transmit}) = \mathbf{0.008}$ or **0.8%**

Answer (d):

- Probability all 3 users transmit = **0.008**
 - Fraction of time queue grows = **0.008** (or **0.8%**)
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3. Suppose Host A wants to send a large file to Host B. The path from Host A to Host

B has three links, of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps.

- a) Assuming no other traffic in the network, what is the throughput for the file transfer?**
- b) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?**
- c) Repeat (a) and (b), but now with R_2 reduced to 100 kbps.**

Given:

- Host A to Host B path has 3 links:
 - $R_1 = 500$ kbps
 - $R_2 = 2$ Mbps = 2000 kbps
 - $R_3 = 1$ Mbps = 1000 kbps
 - File size = 4 million bytes = 4×10^6 bytes = 32×10^6 bits
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(a) Throughput for the file transfer

In a multi-link path, The throughput is determined by the slowest (bottleneck) link:

- $R_1 = 500$ kbps
- $R_2 = 2000$ kbps
- $R_3 = 1000$ kbps

Minimum = 500 kbps

Answer (a): Throughput = 500 kbps

(b) Time to transfer 4 million bytes

Convert file size to bits:

- 4 million bytes = 32 million bits = 32×10^6 bits

Time = File size / Throughput
= $(32 \times 10^6 \text{ bits}) / (500 \times 10^3 \text{ bps})$
= 64 seconds

Answer (b): 64 seconds

(c) With R_2 reduced to 100 kbps

New link rates:

- $R_1 = 500$ kbps
- $R_2 = 100$ kbps
- $R_3 = 1000$ kbps

Minimum = 100 kbps

Answer (c-a): Throughput = 100 kbps

Time = $(32 \times 10^6 \text{ bits}) / (100 \times 10^3 \text{ bps})$
= 320 seconds

Answer (c-b): 320 seconds

4.

a) What is self-replicating malware?

b) Describe how a botnet can be created and how it can be used for a DDoS attack.

c) Suppose Alice and Bob are sending packets to each other over a computer network. Suppose Trudy positions herself in the network so that she can capture all the packets sent by Alice and send whatever she wants to Bob; she can also capture all the packets sent by Bob and send whatever she wants to Alice. List some of the malicious things Trudy can do from this position.

a) Self-replicating malware

Self-replicating malware is malicious software that can copy itself automatically and spread across systems without human intervention. Common examples include worms and viruses.

Two common types are:

- **Viruses:** These attach to normal files or programs. When the file is opened, the virus runs and spreads to other files.
- **Worms:** These spread on their own through networks by finding weaknesses in other computers.

In short, this kind of malware spreads by itself and can infect many systems quickly.

b) Describe how a botnet can be created and how it can be used for a DDoS attack.

Creating a botnet:

- A virus or malware is created or used to infect many computers. These infected computers are called **bots** or **zombies**.
- The virus spreads through **fake emails**, **malicious websites**, or **security vulnerabilities** in software.
- Once a computer is infected, it connects to a **Command-and-Control (C&C) server**, which is controlled remotely.
- The controller can then send instructions to all the infected computers at once.

Using a botnet for a DDoS attack:

- The controller instructs all the bots to **send a large number of requests** to a specific target, such as a website or server.
- The target receives **too much traffic** and becomes very slow or may **crash completely**.
- This is called a **Distributed Denial of Service (DDoS)** attack because it is launched **simultaneously from many devices**, making it hard to stop.

c) Malicious actions Trudy can do as a man-in-the-middle

If Trudy is sitting between Alice and Bob on a network, she can see, block, and change all the messages they send to each other. This is called a **Man-in-the-Middle (MitM) attack**, and it can be very dangerous if the communication is not secure.

Here are some harmful things Trudy can do:

1. **Eavesdrop:** She can **secretly read all the messages** between Alice and Bob. This allows her to steal private information like **passwords, credit card numbers, or personal conversations**.
2. **Modify Messages:** Trudy can **change the content** of the messages. For example, if Alice sends money to Bob, Trudy could change the amount or the bank details.
3. **Impersonate (Spoofing):** Trudy can **pretend to be Alice when talking to Bob**, and **pretend to be Bob when talking to Alice**. Neither of them will know they are actually talking to Trudy.
4. **Drop Packets:** She can **block or delete messages** without anyone knowing. This can cause confusion, delay, or make the system stop working properly.
5. **Inject False Data:** Trudy can **send fake messages** to either Alice or Bob. This can lead to wrong actions or even cause a system failure.
6. **Replay Attacks:** She can **record a real message and resend it later** to trick one of them into doing something harmful or unnecessary.

In short, Trudy can spy on, change, or even fake communication between Alice and Bob. These attacks are very serious and show why **encryption and authentication** are important to keep communication safe.

5.

Consider an application that transmits data at a steady rate (for example, the sender generates an N -bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:

a) Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?

b) Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why?

(a) Which network is better: packet-switched or circuit-switched? Why?

A **circuit-switched network** is better for this kind of application.

This is because the application sends data at a **steady and regular rate** and runs for a **long time** without stopping. In circuit switching, a **dedicated path** is created between the sender and receiver before the data starts flowing. This path stays connected the whole time, just like a phone call.

Since the connection is fixed, the application can **send data smoothly without delay**, and there is **no need to compete with other traffic**. This is perfect for continuous data transfer, like a video call or a long file transfer. On the other hand, packet switching shares the network with others, so delays may happen.

So, circuit-switched networks are more reliable for applications that need **constant and steady data flow**.

(b) Is congestion control needed in a packet-switched network in this case? Why or why not?

No, congestion control is not needed in this case.

The question says that **all applications send data at a steady rate**, and the **total data is less than the capacity of the network links**. This means that the network has **enough space and speed** to carry all the data without getting overloaded.

In packet switching, data from different users is broken into packets and sent through the network. Normally, if too many packets are sent at once, the network can get full and slow down. This is called **congestion**. But in this case, since the **network is not full**, and all traffic is steady and predictable, **congestion will not happen**.

So, we don't need any congestion control system because there is **no risk of overload or delay** in the network.

6. Review the car-caravan analogy in Section 1.4. Assume a propagation Speed of 100 km/hour.

a) Suppose the caravan travels 175 km, beginning in front of one tollbooth, passing through a second tollbooth, and finishing just after a third tollbooth. What is the end-to-end delay?

b) Repeat (a), now assuming that there are eight cars in the caravan instead of ten.

Given Information (From the Analogy):

- **Propagation speed** = 100 km/hour
 - **Distance** = 175 km
 - **Tollbooth processing time per car** = 12 seconds
 - In part (a): **10 cars**
 - In part (b): **8 cars**
 - Number of **tollbooths** = 3 (start at first, pass second, end just after third)
 - So, **2 road segments** between tollbooths
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Important Concepts:

In the car-caravan analogy:

- Each **tollbooth** is like a **router** that processes data one unit at a time.
 - **Tollbooth delay** = time taken for all cars to pass through one tollbooth = **number of cars × 12 seconds**
 - **Propagation delay** = time to travel between tollbooths = $\text{distance} \div \text{speed}$
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Part (a): End-to-End Delay for 10-Car Caravan (in Seconds)

Step 1: Propagation Delay

- Time to travel 175 km = $\frac{175 \text{ km}}{100 \text{ km/hour}} = 1.75 \text{ hours}$
- Convert to seconds:

$$1.75 \text{ hours} \times 3600 \text{ sec/hour} = 6300 \text{ seconds}$$

Step 2: Transmission Delay at Tollbooths

- Time to transmit 10 cars = $10 \times 12 \text{ sec} = 120 \text{ sec}$ per tollbooth
- Total for 3 tollbooths = $3 \times 120 = 360 \text{ sec}$

Step 3: Total End-to-End Delay

$$6300 \text{ sec (propagation)} + 360 \text{ sec (transmission)} = 6660 \text{ seconds}$$

Answer:

The end-to-end delay is 6660 seconds.

Part (b): End-to-End Delay for 8-Car Caravan (in Seconds)

Step 1: Propagation Delay (Same as Part a)

- 6300 seconds (unchanged).

Step 2: Transmission Delay at Tollbooths

- Time to transmit 8 cars = $8 \times 12 \text{ sec} = 96 \text{ sec}$ per tollbooth
- Total for 3 tollbooths = $3 \times 96 = 288 \text{ sec}$

Step 3: Total End-to-End Delay

$$6300 \text{ sec} + 288 \text{ sec} = 6588 \text{ seconds}$$

Answer:

The end-to-end delay is 6588 seconds.

7. This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

- a) Express the propagation delay, d_{prop} , in terms of m and s .
- b) Determine the transmission time of the packet, d_{trans} , in terms of L and R .
- c) Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.
- d) Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{\text{trans}}$, where is the last bit of the packet?
- e) Suppose d_{prop} is greater than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- f) Suppose d_{prop} is less than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- g) Suppose $s = 2.5 \times 10^8$, $L = 1500$ bytes, and $R = 10$ Mbps. Find the distance m so that d_{prop} equals d_{trans} .

Understanding the Basics

Propagation Delay: How long it takes for a single bit to travel from sender to receiver through the physical medium (like a wire or fiber optic cable).

Transmission Delay: How long it takes to push all the bits of a packet onto the transmission medium.

Think of it like a train analogy:

- **Propagation delay** = How long the train takes to travel from station A to station B
- **Transmission delay** = How long it takes to load all the cars onto the track at station A

Solutions

a) Propagation Delay Expression

The propagation delay is the time it takes for a bit to travel from Host A to Host B.

$$d_{\text{prop}} = m/s$$

Where:

- m = distance between hosts (meters)
- s = propagation speed (meters/second)

Simple explanation: If you need to travel m meters at s meters per second, it takes m/s seconds.

b) Transmission Time Expression

The transmission time is the time it takes to push all L bits of the packet onto the link at rate R bps.

$$d_{\text{trans}} = L/R$$

Where:

- L = packet size (bits)
- R = transmission rate (bits per second)

Simple explanation: If you have L bits to send at R bits per second, it takes L/R seconds to send all bits.

c) End-to-End Delay Expression

End-to-end delay is the total time from when the packet starts being sent until it fully arrives at the destination. It includes both the transmission delay (time to push the bits into the link) and the propagation delay (time for bits to travel through the link)

If we ignore processing and queuing delays, the total delay is just the sum of propagation delay and transmission delay.

$$\text{Total delay} = d_{\text{prop}} + d_{\text{trans}} = m/s + L/R$$

Simple explanation: The total time = time to send all bits + time for bits to travel across the link.

d) At time $t = d_{\text{trans}}$, where is the last bit of the packet?

- At $t=0$ Host A starts transmitting the first bit of the packet.
- At $t = d_{\text{trans}}$, the last bit of the packet has just been pushed onto the link.
- The last bit is now at the **beginning of the link**, just starting its propagation to Host B.

Answer:

The last bit is **on the link, just leaving Host A**.

e) Location of the First Bit at $t = d_{\text{trans}}$ (When $d_{\text{prop}} > d_{\text{trans}}$)

If the propagation delay is greater than the transmission delay ($d_{\text{prop}} > d_{\text{trans}}$), the first bit, which started propagating at $t=0$, has been traveling for d_{trans} seconds but has not yet reached Host B. Thus, at $t = d_{\text{trans}}$, the first bit is **somewhere along the link**, still moving toward Host B.

f) Location of the First Bit at $t = d_{\text{trans}}$ (When $d_{\text{prop}} < d_{\text{trans}}$)

If the propagation delay is less than the transmission delay ($d_{\text{prop}} < d_{\text{trans}}$), the first bit would have already arrived at Host B by $t = d_{\text{prop}}$. Since $d_{\text{prop}} < d_{\text{trans}}$, at $t = d_{\text{trans}} = d_{\text{trans}}$, the first bit has **already reached Host B**, while the last bit is still being transmitted.

g) Given:

- $s = 2.5 \times 10^8$ meters/sec
- $L = 1500$ bytes = $1500 \times 8 = 12000$ bits
- $R = 10$ Mbps = 10×10^6 bps

We are to find distance m such that $d_{prop} = d_{trans}$:

$$d_{prop} = d_{trans} \Rightarrow \frac{m}{s} = \frac{L}{R} \Rightarrow m = \frac{L}{R} \times s$$

Substitute values:

$$m = \frac{12000}{10 \times 10^6} \times 2.5 \times 10^8 = 0.0012 \times 2.5 \times 10^8 = 3 \times 10^5 \text{ meters} = \boxed{300 \text{ km}}$$

8.

Two hosts, A and B, are connected by a direct communication link with a transmission rate of 5 Mbps. They are separated by a distance of 20,000 kilometers, and the propagation speed over the link is 2.5×10^8 meters/second.

- Calculate the bandwidth-delay product of the link.
- Suppose Host A sends a file of 800,000 bits to Host B as a continuous, single message. What is the maximum number of bits that can be present in the link at any given time during the transmission?
- Provide an interpretation of the bandwidth-delay product.
- What is the width (in meters) of a bit in the link? Is it longer than a football field?
- Derive a general expression for the width of a bit in terms of the propagation speed s , the transmission rate R , and the length of the link m .

Given:

Transmission rate $R = 5 \text{ Mbps} = 5 \times 10^6 \text{ bits/s}$

Distance $m = 20,000 \text{ km} = 2.0 \times 10^7 \text{ m}$

Propagation speed $s = 2.5 \times 10^8 \text{ m/s}$

a) Bandwidth–delay product (BDP)

Propagation delay:

$$t_p = \frac{m}{s} = \frac{2.0 \times 10^7}{2.5 \times 10^8} = 0.08 \text{ s (80 ms)}$$

BDP:

$$\text{BDP} = R \cdot t_p = (5 \times 10^6) \cdot 0.08 = 4.0 \times 10^5 \text{ bits}$$

Answer: 400,000 bits ($\approx 50,000$ bytes ≈ 48.8 KiB).

b) Maximum number of bits in the link during transmission.

Host A sends a file of 800,000 bits as a continuous single message. The maximum number of bits that can be in the link at any time is limited by the bandwidth-delay product (if the file is larger than BDP). Since the file size (800,000 bits) is larger than the BDP (400,000 bits), the maximum bits in the link is the BDP itself.

Why? Because after transmitting for one propagation delay, the first bit reaches the receiver, but the sender has transmitted $R \times \tau$ bits without acknowledgment. So the link can hold at most BDP bits.

File size 800,000 bits > BDP 400,000 bits

Answer: 400,000 bits.

c) Interpretation of the bandwidth-delay product.

The bandwidth-delay product represents the maximum amount of data (in bits) that can be "in flight" over the link at any given time. It is the product of the link's capacity (bandwidth) and the end-to-end propagation delay. This is also known as the "pipe size" of the link, indicating how many bits must be transmitted to fully utilize the link.

d) Width of a bit in the link

The width of a bit is the **physical length** occupied by one bit on the transmission medium.

1. Time to transmit one bit:

$$T_{\text{bit}} = \frac{1}{R} = \frac{1}{5 \times 10^6} = 2 \times 10^{-7} \text{ seconds}$$

2. Distance propagated in this time:

$$\text{Bit width} = s \times T_{\text{bit}} = (2.5 \times 10^8) \times (2 \times 10^{-7}) = 50 \text{ meters}$$

A football field is about **100 meters long**. So one bit (50 m) is **half the length of a football field**.

Answer: 50 meters, shorter than a football field.

e) General expression for the width of a bit

Let:

- s = propagation speed (m/s)
- R = transmission rate (bits/s)
- m = length of the link (m)

1. Time to transmit one bit:

$$T_{\text{bit}} = \frac{1}{R}$$

2. Distance propagated in this time:

$$\text{Bit width} = s \times T_{\text{bit}} = \frac{s}{R}$$

This shows that the bit width is independent of link length m .

It only depends on propagation speed and transmission rate.

Answer:

$$\text{Bit width} = \frac{s}{R}$$

9.

a) What information is used by a process running on one host to identify a process running on another host?

b) Recall that TCP can be enhanced with TLS to provide process-to-process security services, including encryption. Does TLS operate at the transport layer or the application layer? If the application developer wants TCP to be enhanced with TLS, what does the developer have to do? 10

c) What is meant by a handshaking protocol?

d) What does a stateless protocol mean? Is IMAP stateless? What about SMTP?

e) How can websites keep track of users? Do they always need to use cookies?

a) What information is used by a process running on one host to identify a process running on another host?

To identify a process on another host, a combination of the **destination IP address** and **port number** is used.

- The **IP address** identifies the host (computer or server).
- The **port number** identifies the specific process or application running on that host. Together, they form a **socket** (IP address + port number) used for communication.

b) Does TLS operate at the transport layer or the application layer? What must an application developer do to use TLS with TCP?

TLS (Transport Layer Security) does **not operate directly at the transport layer**, but rather **between the application and transport layers**. It is often considered to work at the **application layer** from a developer's point of view.

To use **TLS with TCP**, the application developer must:

- Use a **TLS-enabled library or API**, such as OpenSSL, or use protocols like **HTTPS** (which is HTTP over TLS).
- Modify the application to **initiate and handle TLS handshakes**, encryption, and decryption.

The developer cannot rely on plain TCP sockets alone; they need to use a secure socket (e.g., SSL_socket or HTTPS libraries).

c) What is meant by a handshaking protocol?

A **handshaking protocol** is a communication process where two devices or applications **exchange control messages** before actual data transmission begins.

This handshake ensures:

- Both parties are **ready and willing to communicate**.
- **Connection parameters** are agreed upon (e.g., protocol version, encryption, sequence numbers).

An example is the **three-way handshake** in TCP, used to establish a reliable connection between sender and receiver.

d) What does a stateless protocol mean? Is IMAP stateless? What about SMTP?

A **stateless protocol** is a protocol in which the server does **not store any information** (or "state") about past client requests. Each request is treated as **independent and unrelated** to any previous ones.

- **IMAP (Internet Message Access Protocol)** is **not stateless**. It maintains state about the user's session, such as selected mailbox, read/unread messages, etc.
- **SMTP (Simple Mail Transfer Protocol)** is also **not stateless**. It keeps state during a mail transaction (sender, recipient, message data) until the message is fully delivered.

So, both IMAP and SMTP are **stateful protocols**.

e) How can websites keep track of users? Do they always need to use cookies?

Websites can keep track of users through several methods:

- **Cookies:** Small pieces of data stored in the user's browser, commonly used for login sessions, preferences, and tracking.
- **Session IDs in URLs:** Some sites append session identifiers directly to URLs.
- **Hidden form fields:** State can be passed back to the server via hidden fields in forms.
- **Local storage or session storage:** Modern browsers provide local storage mechanisms similar to cookies.
- **IP address or device fingerprinting:** Used for tracking but less reliable and more privacy-invasive.

So, websites **do not always need to use cookies**, but cookies are the most common and user-friendly method for maintaining state.

10.

- Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?
 - Suppose Alice, with a Web-based e-mail account (such as Hotmail or Gmail), sends a message to Bob, who accesses his mail from his mail server using IMAP. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts.
 - What is the HOL blocking issue in HTTP/1.1? How does HTTP/2 attempt to solve it?
 - What is the difference between recursive and iterative DNS queries?
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(a) How does Web caching reduce delay? Does it reduce delay for all objects? Why or why not?

Web caching stores copies of previously accessed web content (like images, pages, or videos) on a **local cache**—such as a browser cache or a caching proxy server.

How it reduces delay:

- When a user requests a web object (e.g., a page or image), the cache is checked first.
- If the object is **already stored in the cache (a cache hit)**, it is delivered **much faster** than downloading it again from the original server.
- This reduces **loading time, saves bandwidth, and decreases network traffic**.

Does it reduce delay for all objects?

- **No**, only for **some objects**—those that were **previously requested and stored** in the cache.
- If the object is **not in the cache (a cache miss)**, the browser must still fetch it from the original web server.

Summary: Web caching reduces delay for cached objects, but not for new or changed objects that aren't stored in the cache yet.

(b) How does Alice's email reach Bob's host (using webmail and IMAP)-

Let's follow the steps involved:

1. **Alice composes an email** in her webmail (e.g., Gmail or Hotmail) using a browser. This is done over **HTTP/HTTPS**.
2. When she clicks "send", her email is transferred from the **webmail server** to **Bob's mail server** using the **SMTP (Simple Mail Transfer Protocol)**.
 - **SMTP** is the standard protocol for **sending email** between servers.
3. **Bob uses IMAP (Internet Message Access Protocol)** to **access his inbox** from his device (like a phone or laptop).
 - IMAP allows Bob to view and manage his emails without downloading them permanently.
 - Bob's device contacts his mail server using **IMAP**, typically over port 143 (or 993 for secure IMAP).

Protocols used:

- **HTTP/HTTPS** (Alice's browser to webmail server)
- **SMTP** (webmail server to Bob's mail server)
- **IMAP** (Bob's mail server to Bob's device)

(c) What is HOL blocking in HTTP/1.1? How does HTTP/2 solve it?

HOL (Head-of-Line) Blocking in HTTP/1.1:

- In HTTP/1.1, **only one request can be active on a connection at a time.**
- If one request is **delayed or slow**, it **blocks** the next ones from being processed.
- This causes a **queue**, slowing down page loading even if other responses are ready. this is called **Head-of-Line (HOL) blocking**.

How HTTP/2 solves it:

- HTTP/2 uses **multiplexing**, which allows **multiple requests and responses to be sent at the same time over a single connection.**
- So even if one stream is slow, others can continue freely.
- This **removes head-of-line blocking** at the application level and greatly improves performance.

(d) Difference between recursive and iterative DNS queries

Point	Recursive DNS Query	Iterative DNS Query
Who does the work?	The DNS server handles everything.	The client (or resolver) does step-by-step work.
Number of requests	Only one request from the client.	Multiple requests —one for each step (root → TLD → final server).
Response type	Gives the final IP address directly.	Gives hints (like "ask this next server").
Client effort	Low —just waits for the answer.	High —keeps sending new requests based on responses.
Usage	Used by regular devices (PC, phone, browser).	Used between DNS servers to find info.
Speed	Can be slower (server does more work).	Can be faster (work is spread out).
Example	You ask once → DNS server gives the full answer.	You ask the root → it says "ask .com" → .com says "ask example.com".